

Are phosphodiesterase type 5 inhibitors associated with vision-threatening adverse events? A critical analysis and review of the literature.

INTRODUCTION: Phosphodiesterase type 5 (PDE5) inhibitors are the first line drugs for treatment of erectile dysfunction. Sildenafil (Viagra(R)), tadalafil (Cialis(R)), and vardenafil (Levitra(R)) are from the same class of drugs that inhibit PDE5. Transient visual symptoms such as change in color perception and increased light sensitivity are well-known adverse effects of these drugs and occur in 3-11% of sildenafil users. Vision-threatening (serious) ocular complications, such as nonarteritic ischemic optic neuropathy and cilio-retinal artery occlusion have rarely been reported in PDE5 inhibitor users. **AIMS:** To highlight and analyze the most recently published case literature on serious ocular complications of PDE5 inhibitors. **METHODS:** Search of the peer-reviewed English literature was conducted using Medline. The following databases also were searched: Cumulative Index to Nursing and Allied Health Literature, Cochrane Library, Global Health, and MD Consult. The causality assessment of the reported adverse drug reactions was analyzed by applying both the World Health Organization (WHO) Probability Scale and the criteria utilized by the National Registry of Drug-Induced Ocular Side Effects. **MAIN OUTCOME MEASURES:** To scientifically and objectively find out if PDE5 inhibitors are associated with vision-threatening ocular complications. **RESULTS:** Eight case reports of serious PDE5 inhibitor-associated ocular complications were identified since January 2006 until February 2011. Case reports included cases of anterior and posterior nonarteritic ischemic optic neuropathy, central retinal vein occlusion, cilio-retinal artery occlusion, acute angle closure glaucoma and optic atrophy after sildenafil use. **CONCLUSION:** There is lack of conclusive evidence to indicate a direct cause-effect relationship between PDE5 inhibitor use and vision-threatening ocular events. Men who use PDE5 inhibitors appear to suffer vision-threatening complications at the same frequency as the general population. However, minor visual adverse effects occur in 3-11% of users and they are transient and reversible. © 2011 International Society for Sexual Medicine.